



Selective and Cost-Aware Retrieval Routing for RAG

蔡尚融 李羿昕 李政緯

National Yang Ming Chiao Tung University

Abstract

This project studies **retrieval-stage control for RAG** as an offline contextual bandit. For each query, a lightweight policy observes lexical and retrieval-confidence features, then chooses a retrieval or rewrite action. The final claim is intentionally narrow: improved **cost-aware evidence retrieval**, not generated-answer EM/F1 and not LLM fine-tuning.

Problem and Objective

Fixed RAG retrieval cannot adapt to query difficulty. Rewrites, dense retrieval, hybrid retrieval, or LLM experts can help, but only when their expected retrieval gain justifies cost.

- Learn a per-query retrieval/rewrite router from offline rewards.
- Keep cost explicit through rewrite and retrieval-call penalties.
- Validate claims with baselines, bootstrap intervals, OPE, and artifact checks.

Contextual-Bandit Formulation

Each query is a one-step decision:

$$\pi(s) = \arg \max_a \mathbb{E}[R \mid s, a]$$

$$R = \text{Recall@5} + 0.5 \cdot \text{MRR} - \text{RewriteCost} - \text{ExtraRetrievalCallCost}$$

- State:** query features + BM25 confidence statistics.
- Actions:** BM25, dense, hybrid, keyword rewrite; Gemini actions are expensive extensions.
- Policy:** direct-method action-value models with CV model selection.
- Training evidence:** replayable full-information retrieval rewards; this is not online bandit learning, and selected-action replay is a diagnostic baseline.

Implementation Pipeline

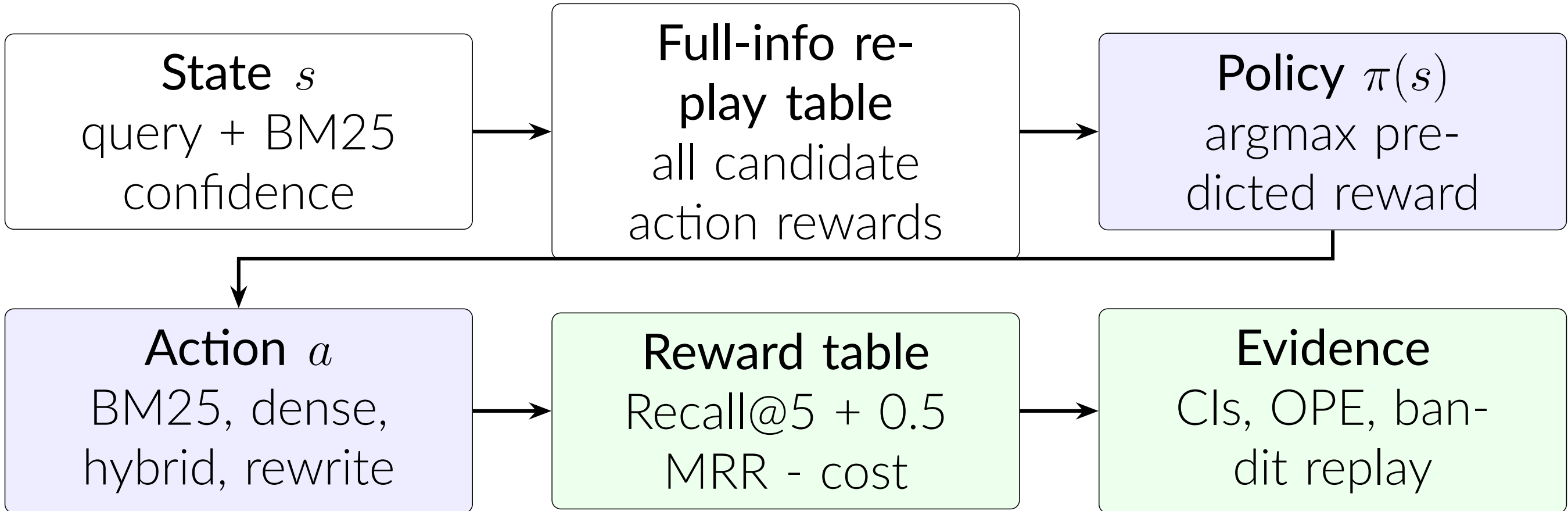
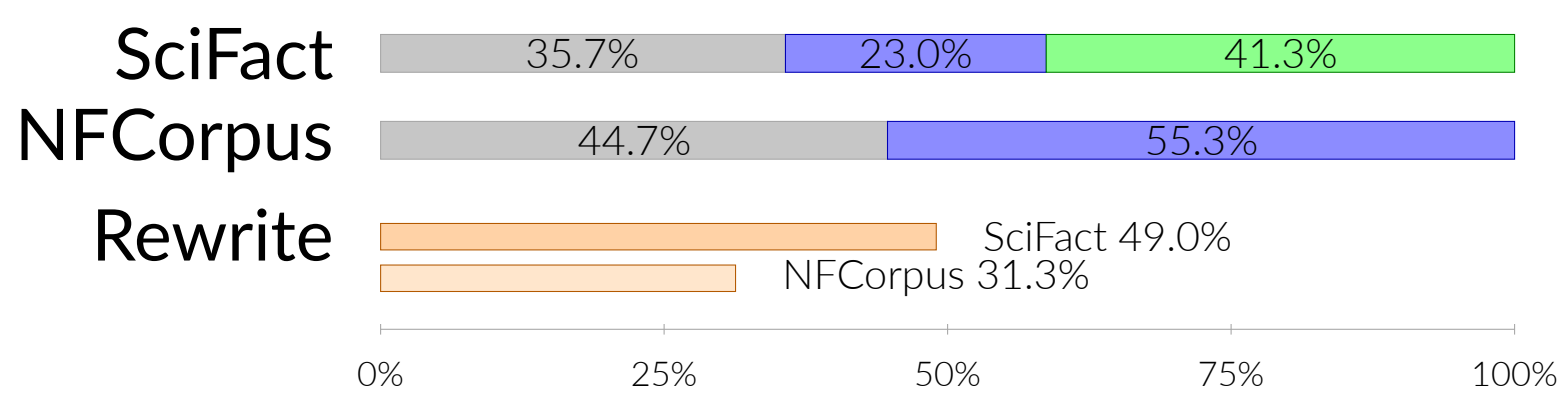


Figure 1. Offline contextual-bandit framing: replayable action rewards train a policy that maps query state to retrieval action.

Policy Behavior



Evidence: `outputs/results/*_retrieval_policy_action_distribution.csv`.

SciFact uses hybrid/rewrite often; NFCorpus stays one-call with BM25 or dense retrieval.

Cross-Dataset Generalization

Evaluating policy transferability by testing trained routing policies on unseen target domains:

Source	Target	Policy Reward	Target Train-Best	Calls
SciFact	SciFact	1.093	1.057	1.477
SciFact	NFCorpus	0.386	0.377	1.597
NFCorpus	NFCorpus	0.407	0.377	1.000
NFCorpus	SciFact	1.064	1.057	1.000

The NFCorpus-trained policy transfers to SciFact with positive reward delta over target train-best. The SciFact-trained policy transfers to NFCorpus but incurs extra retrieval cost.

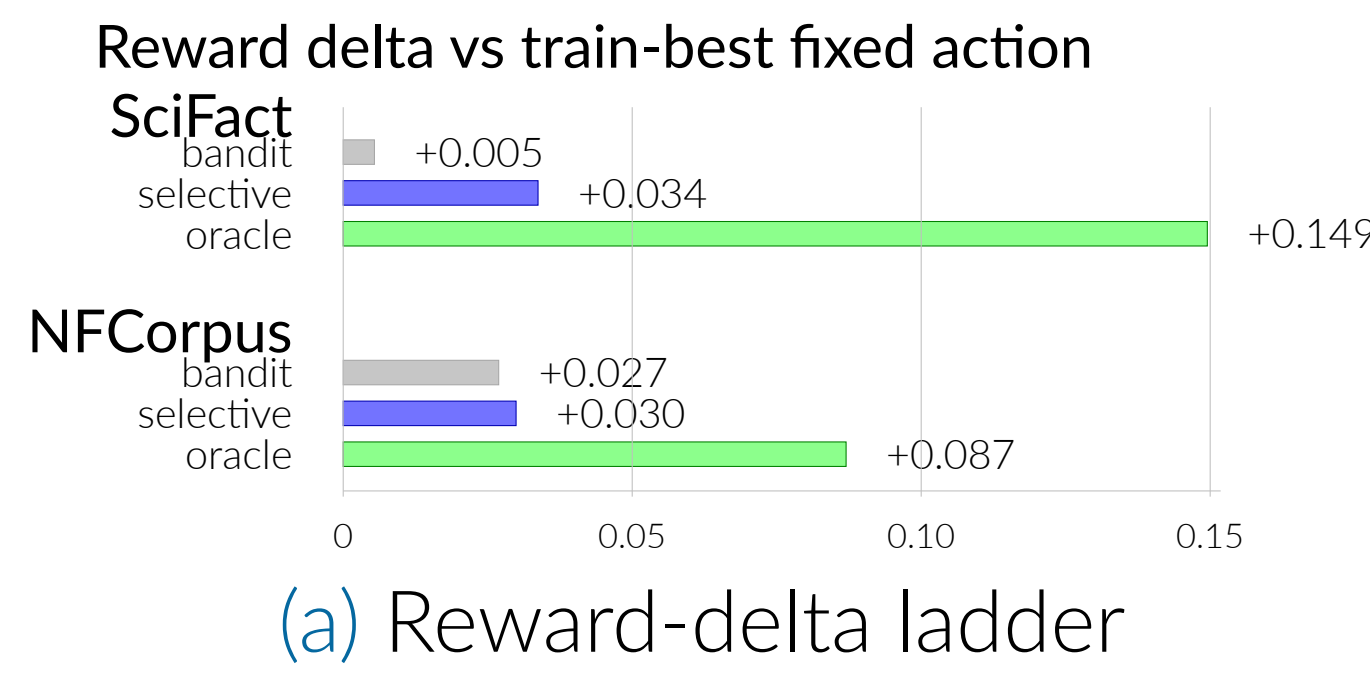
Datasets and Evidence Boundary

Final claims use seed-42 full-corpus BEIR runs with 600 train queries and 300 test queries per dataset: **SciFact** scientific-claim retrieval over 5,183 documents and **NFCorpus** biomedical/nutrition retrieval over 3,633 documents. Smoke tests verify code paths only.

Main Results

Table 1. Full-corpus BEIR evidence from `outputs/results/final_main_results_table.csv`. Reward rows use the train-best reward baseline. Constrained rows report utility value and delta vs the train-best utility baseline at $\lambda = 0.03$: SciFact 1.0420, NFCorpus 0.3772.

Dataset	Method	Obj.	Value	Δ basis / 95% CI	Calls
SciFact	Train-best fixed	reward	1.0568	0.0000 vs reward base	2.000
SciFact	Selected-action bandit	reward	1.0621	+0.0054 vs reward base	1.033
SciFact	Selective policy	reward	1.0905	+0.0337 vs reward base [0.0089, 0.0583]	1.413
SciFact	Constrained $\lambda = 0.03$	utility	1.0894	+0.0474 vs utility 1.0420 [0.0143, 0.0822]	1.273
SciFact	Oracle action	reward	1.2062	+0.1495 vs reward base	1.007
NFCorpus	Train-best fixed	reward	0.3772	0.0000 vs reward base	1.000
NFCorpus	Selected-action bandit	reward	0.4040	+0.0269 vs reward base	1.137
NFCorpus	Selective policy	reward	0.4071	+0.0299 vs reward base [0.0054, 0.0548]	1.000
NFCorpus	Constrained $\lambda = 0.03$	utility	0.4098	+0.0327 vs utility 0.3772 [0.0106, 0.0584]	1.000
NFCorpus	Oracle action	reward	0.4641	+0.0870 vs reward base	1.020



(a) Reward-delta ladder

Defense Numbers

Metric	SciFact	NFCorpus
Selective reward Δ	+0.0337	+0.0299
Gap closed	22.6%	34.4%
Constrained utility Δ	+0.0474	+0.0327
Selective calls/query	1.413	1.000
3-seed win	0.667	0.667

CI evidence: reward CIs stay above zero on both datasets; utility CIs are +0.0143 to +0.0822 and +0.0106 to +0.0584.

(b) Numeric readout

Figure 2. Primary visual evidence: selective-policy gains plus readable cost-aware and robustness diagnostics.

The learned policy improves reward over the train-selected fixed action on both BEIR datasets, closing 22.6% of the SciFact oracle gap and 34.4% of the NFCorpus oracle gap. The constrained policy keeps positive utility gains while making call tradeoffs explicit.

Diagnostics That Support the Claim

- Selected-action replay:** Linear Thompson delta is +0.005 on SciFact and +0.027 on NFCorpus.
- Replay regret:** LinUCB selected-feedback regret is 82.1 vs 67.4 full-information direct on SciFact, and 43.2 vs 29.6 on NFCorpus.
- OPE coverage warning:** SciFact IPS error rises from 0.127 under uniform logging to 0.268 under sparse train-best logging.
- Three-seed robustness:** selective win rate is 0.667 on both datasets; mean deltas are +0.021 on SciFact and +0.025 on NFCorpus.
- Audit trail:** markdown consistency, evidence consistency, checkpoint manifest, and artifact index pass.

Where the Result Gets Stronger

- Difficulty buckets:** high action-disagreement queries show larger gains: SciFact +0.049 reward, NFCorpus +0.096 reward.
- Budget sweep:** SciFact keeps +0.039 utility at 1.047 calls/query with $\lambda = 0.06$; NFCorpus keeps +0.033 utility at 1.000 calls/query.
- Gate baseline:** a margin gate adds only +0.0006 over selective on SciFact and is not better on NFCorpus.

Evidence: `outputs/results/poster_improvement_summary.csv`.

These checks turn the next-step analysis into evidence: the router is strongest on high-disagreement query slices, constrained routing can preserve utility near one call/query, and simple margin gating does not materially improve the main policy.

Limitations and Claim Boundary

- Evidence retrieval only; no generated-answer EM/F1 or LLM fine-tuning.
- Direct method uses replayable full-information action tables.
- Not online bandit learning; OPE is simulated, not real online logs; semantic features remain analysis-only.